THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following reverses the force on a current-carrying conductor in a field? 1

(a) Reducing the length **(b)** Reversing the current
(c) Heating it **(d)** Increasing the current

2. Direction of the induced current is obtained from 1

(a) Ohm's law **(b)** Fleming's left-hand rule
(c) Fleming's right-hand rule **(d)** Joule's law

3. A solenoid is wound over a soft iron rod. This arrangement is 1

(a) an electromagnet **(b)** a resistor
(c) a permanent magnet **(d)** a fuse

4. When both the current and the field are reversed, the force 1

(a) is reversed **(b)** doubles
(c) remains the same direction **(d)** becomes zero

5. In a generator the current changes direction after every 1

(a) full rotation **(b)** four rotations **(c)** half rotation **(d)** two rotations

6. Which core is used for the electromagnet in an electric bell? 1

(a) Steel **(b)** Soft iron **(c)** Brass **(d)** Copper

7. A positively charged particle moves into a magnetic field directed into the page, moving to the right. The force is 1

(a) into the page **(b)** to the left
(c) upward (on the page) **(d)** downward

8. The frequency of AC is 50 Hz. The time period of AC is 1

(a) 0.2 s **(b)** 5 s **(c)** 50 s **(d)** 0.02 s

9. Steel is not used for making an electromagnet because 1

(a) it melts easily
(b) it retains magnetism after the current stops
(c) it is a nonmetal
(d) it is expensive

10. A moving charge in a magnetic field, moving perpendicular to the field, follows a path that is 1

(a) straight **(b)** circular **(c)** zigzag **(d)** parabolic

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the galvanometer connected to a coil show no deflection when the magnet is held at rest inside the coil? 2

12. Why is the wire of a fuse made of a material having a low melting point? 2

13. Why is the compass needle deflected more when kept nearer to a wire? 2

14. Why does a generator's output become AC? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a current-carrying coil kept near a compass needle deflects it, and what happens when the current is reversed. 3

16. What are the conditions for the induced current in a closed coil? 3

17. State the energy conversions taking place in a generator and electromagnet. 3

OR

A student winds 100 turns of insulated copper wire on an iron nail and connects it to a battery. What will happen to iron pins brought near? How can she make it stronger? 3

18. Describe the colour coding of wires and the voltage between live and neutral wires in India. 3

OR

If the speed of rotation of the coil of a generator is increased, what happens to the output? Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A straight conductor carrying current of 5 A is placed in a magnetic field. Explain the force on it with Fleming's left-hand rule. If the current flows north to south and the field is vertically upward, find the direction of force. 5

OR

Distinguish between direct current and alternating current. State the frequency in India, the period, and the number of reversals per second. 5

20. Describe an experiment to demonstrate the force on a current-carrying rod placed in the field of a horseshoe magnet. State two ways to change its direction. 5

OR

Explain with a labelled description of the solenoid how we can find the polarity of its ends using the right-hand rule. How does a solenoid act as a bar magnet? 5

END OF QUESTION PAPER — Total: 40 Marks